

Transfusion for paediatric patients in Children's Emergency

1.1 Paediatric patients in haemorrhagic shock due to ongoing actively bleeding require urgent and timely blood transfusion, as well as prompt control of the bleeding source.

1.2 At NUH, the initial pack of Massive Transfusion Protocol (MTP1) can be activated for patients >40kg **or** >12 year old, provided they require the complete set of blood products to adequately replace their blood loss and to avoid unnecessary wastage of precious blood resources.

MTP 1 consists of 4 units of PCT, 4 units of FFP, and 1 unit of pooled platelets / 1 unit CSP.

1.3 This table summarizes the different types of crossmatch:

For Red Blood Cells				
Type of Crossmatch	Time from ordering or if urgent, calling BTS to first 2 PCT ready for collection*	GXM required?	ABO group	Transfusion Reaction Risk
<u>Uncrossmatch</u>	5 – 10 mins	No. Can be issued before GXM**	O+ for males O- for Indian and Caucasian females <50 y	5% risk of incompatibility 0.5% risk of <u>haemolytic</u> reaction
Rapid Match	15min from time GXM sample arrives in BTS	Yes	ABO specific No antibody screen	5% risk of incompatibility 0.5% risk of <u>haemolytic</u> reaction
Full Crossmatch	15min (valid GXM in last 3 days, without antibody) 45min from the time GXM sample arrives in BTS 2-4 hours (with antibody)	Yes	ABO specific Antibody screen done	Lowest risk of <u>haemolytic</u> reaction

The following outlines the various terms used for obtaining blood products from the blood bank:

a) Uncrossmatched blood –

What we need to know
-This is reserved for patients in haemorrhagic shock with exsanguinating haemorrhage, whereby blood product is delivered to patient within 10 minutes, even before the GXM is processed. -We can tap on EMD blood fridge resources for this group of patients. -The blood given would be O+ blood for all patients (which would be available in the EMD blood fridge), except those with known O- blood group or those with unknown blood group and are females of reproductive age group of Indian/Caucasian descent, for which the O- blood product would need to be obtained from the blood bank.
What we need to do
-The ordering clinician must sign a form to acknowledge the risk of haemolysis during the blood order for E-blood on EPIC, or on a manual form during EPIC downtime (see Appendix I). -The EMD MTP nurse needs to be informed about the use of EMD blood fridge resources. -Blood products must only be retrieved one pack at a time. -There is NO return of the blood product back into the fridge once it has been removed. -If O- blood is required, blood bank needs to be informed immediately with plans made to collect blood product from blood bank. -A GXM should be sent before transfusion occurs.

b) Rapid crossmatch blood –

What we need to know
-This is intended for patients who display early signs of haemorrhagic shock but can wait at least 30-45 minutes for blood to arrive. -Blood bank performs a type and screen within 15 minutes upon receiving GXM sample.
What we need to do
-A GXM sample must be drawn, ordered as STAT, labelled and sent to the lab. -The clinician must contact blood bank (22305) to notify them of the incoming GXM sample for rapid crossmatch and to request for blood collection. -The team must send a member or arrange a porter to bring the patient's identification label to the blood bank (main building level 3) to collect the blood product 15 minutes after GXM has been received by blood bank. -The order in EPIC should be placed under BTS blood order and must be entered as STAT, not routine (see appendix II for order details). -A manual form to acknowledge risk of haemolysis is signed and submitted to blood bank.

c) Full crossmatch blood

What we need to know
-This is for haemodynamically stable patients who need a non-urgent transfusion. -Blood bank performs type and screen in 45 minutes upon receiving GXM sample. -It can take up to 2-4 hours if antibodies are detected in the GXM sample.
What we need to do
-A GXM sample must be drawn and sent to the lab. -The transfusion can be deferred to when the patient is admitted in the ward, unless the patient is not getting admitted (e.g AOR discharge) or there is a significant delay in admission. -The order in EPIC should be placed under BTS blood order and is entered as routine. (see appendix II for order details).

Use of EMD blood fridge (EMBF) products for Paediatric patients in haemorrhagic shock (w.e.f. 15 March 2026)

2.1 Usage of EMBF blood products is permitted for paediatric patients in haemorrhagic shock, including non-trauma cases. This usage should follow the principles of balanced haemostatic resuscitation, with a transfusion ratio of 1:1:1 (PCT:FFP:Platelets) for paediatric trauma patients, and an individualised approach without a fixed ratio in non-trauma patients. Activation of MTP should be considered for children who meet institutional MTP criteria (see ref 1.2). For patients who do not meet the criteria, FFP and platelets should be ordered individually from the blood bank while maintaining the recommended balanced transfusion approach.

2.2 The EMBF consists of 4 units of O+ packed cells, of which 1 unit is a Leucocyte-reduced packed cell. The leucocyte-reduced packed cell is particularly beneficial for the following groups of patients:

1. Children <1 year old
2. Those at risk of febrile haemolytic reactions or risk of CMV transmission (e.g. immunocompromised patients)

While especially indicated for these groups, leucocyte-reduced packed cells may be administered to any patient if clinically required.

2.3 Contraindications to use of EMBF products (O+ PCT):

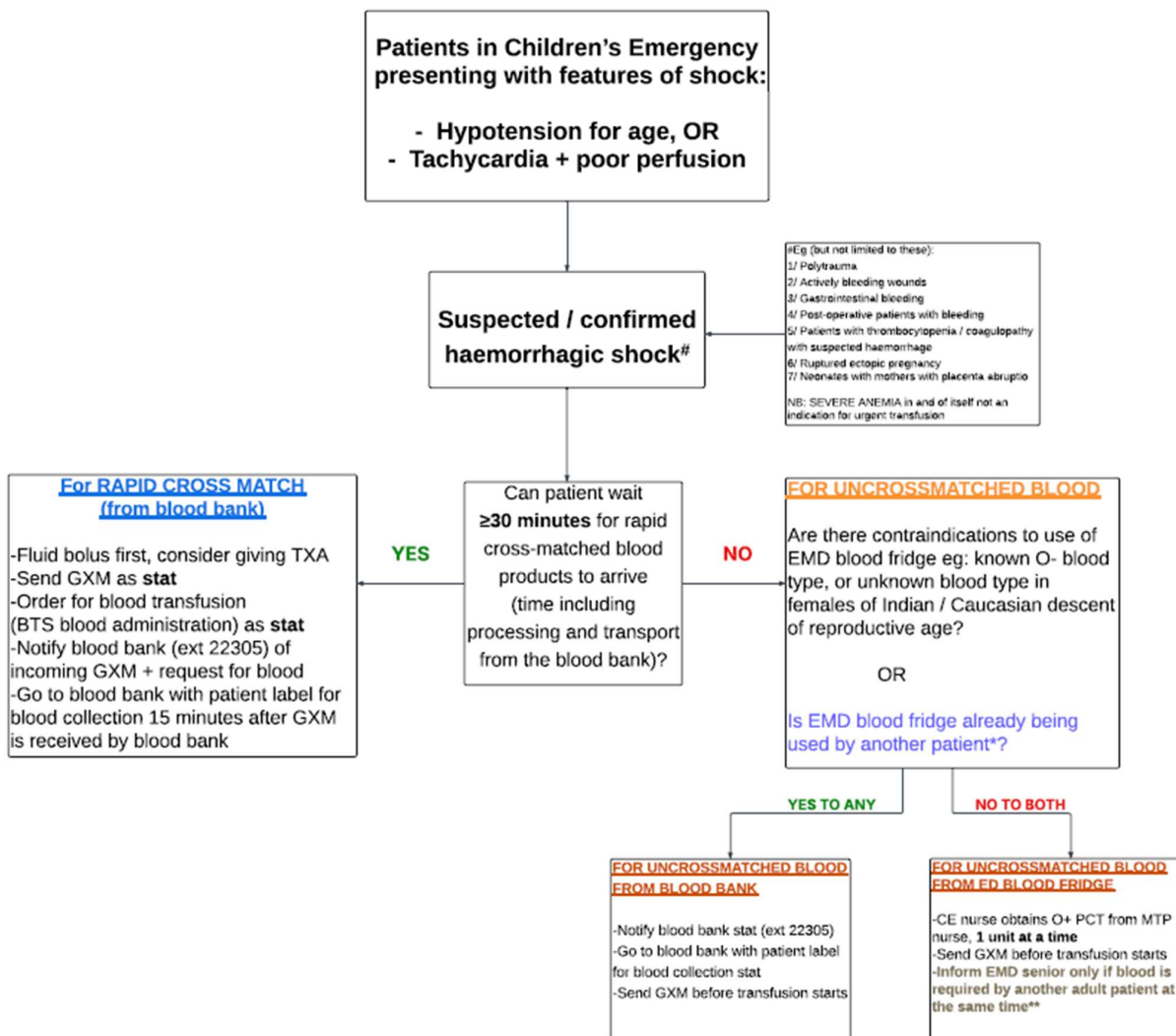
1. Patients known to be O- blood group
2. Females of reproductive age (Indian or Caucasian descent) with unknown blood group

2.4 In Paediatric trauma patients exhibiting signs of haemorrhagic shock requiring urgent blood transfusion, the Trauma Team Leader and Children's Emergency senior can ask for EMBF packed cells to be given. Guide on withdrawal of EMBF products, based on age / weight:

- Age $\leq$ 1yr or weight <10kgs: Withdraw 1 unit of leucocyte-reduced packed cells
- Age 1 – 12yr and weight <40kg: Withdraw 1 unit of packed cells
- Age > 12yr or weight >40kg: Withdraw 1 unit of packed cells and check with the Trauma Team Leader if MTP is to be activated

2.5.1 In other cases of paediatric haemorrhagic shock (ie non-trauma) where urgent blood transfusion is required and rapid crossmatched blood is not available in time, **the Children's Emergency senior can request for EMBF packed cells to be given after ensuring no contraindications (patients with known O- blood group).**

BLOOD TRANSFUSION IN PAEDIATRIC HAEMORRHAGIC SHOCK



* If the EMBF products are already being used for by an adult EMD patient, the CE senior must obtain blood directly from blood bank and request uncrossmatched blood.

** If the EMBF products are required simultaneously for both an adult EMD patient and a paediatric patient, the available EMBF units should be divided (1 unit for the paediatric patient, 3 units for the adult patient).

In such a situation, the CE senior must inform the adult EMD senior, who will then arrange to obtain the remaining unit of PCT from blood bank.

The CE senior should also arrange for subsequent blood products with blood bank.

2.5.2 The decision for the appropriate activation of the EMD fridge blood products (weighing the clinical urgency for transfusion against the risk of uncrossmatched blood products use (ref Table 1.3)) is the sole responsibility of the CE Senior Doctor in charge of the patient.

2.5.3 The adult EMD senior doctors will **not** be involved in the clinical decision for either the usage of EMD Fridge blood products for CE patients, or any subsequent matters related to the transfusion.

The CE clinicians will only need to inform and communicate with the EMD Senior Doctor in the event of a simultaneous EMD and CE need for use of the EMD Fridge blood products (refer 2.5.9).

2.5.4 To obtain the EMBF products, the CE nurse will approach the adult EMD MTP nurse on duty. The EMD MTP nurse will then retrieve the blood product and hand it over to CE nurse. Thereafter, the responsibility for the blood product usage, safe custody and subsequent liaising with the blood bank on patient details, transfusion related processes will be assumed by the CE nurse taking possession of the blood product.

2.5.5 The blood products are to be retrieved **one pack** at a time.

2.5.6 Once removed from the blood fridge, the blood product **cannot** be returned. If it is removed but not transfused, the Return of Blood Product(s) Form needs to be filled (see Appendix III). All incidents of return of unused EMBF products will be investigated and reported.

2.5.7 The blood product must be commenced in CE prior to patient transfer. It must be actively transfusing before departure and should not be issued electively for use at another location (e.g. in OT).

2.5.8 If the EMBF products are already being used for by an adult EMD patient, the CE senior must obtain blood directly from blood bank and request uncrossmatched blood.

2.5.9 If the EMBF products are required simultaneously for both an adult EMD patient and a paediatric patient, the available EMBF units should be allocated as follows: 1 unit for the paediatric patient and 3 units for the adult patient.

In such a situation, **the CE senior must inform the adult EMD senior**, who will then arrange for the additional 1 unit of packed red cells to be obtained from the blood bank. The CE senior must also anticipate the need for further blood products and arrange early retrieval from the blood bank should the paediatric patient require additional blood products after the transfusion of the first unit.

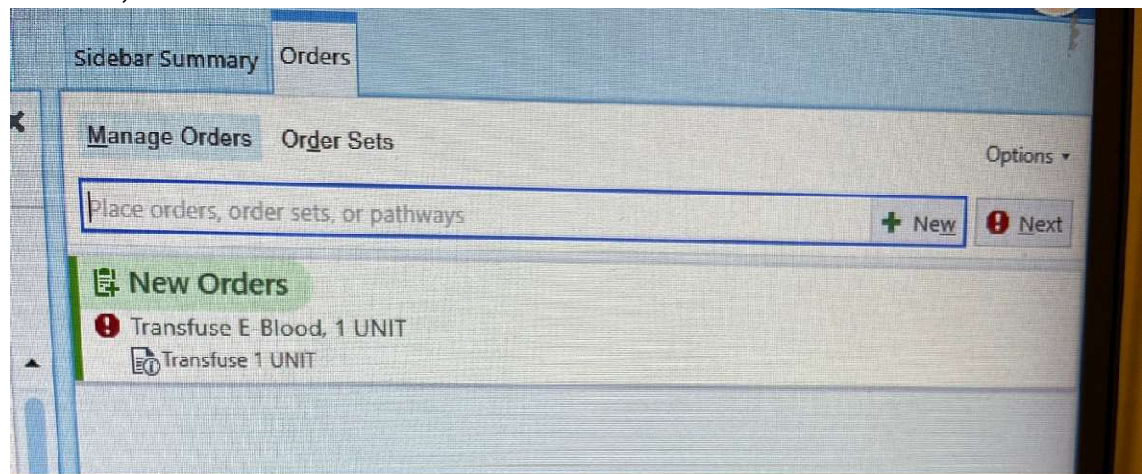
2.5.10 All transfusions performed in CE, whether using blood bank or blood fridge products, must be audited

- **Blood fridge:** Ensure responsible and accountable use of blood products
- **Blood bank:** Ensure timely delivery of blood products to the patient

After any transfusion that has occurred in Children's Emergency, please notify Drs Ivy Ang and Jasmine Ho via Teams. A real-time case review will be conducted, and the blood bank team and the adult ED doctor in charge of EMBF will be emailed to maintain accountability and uphold good governance.

Appendix I – How to order on EPIC blood fridge blood / uncrossmatched blood from blood bank:

1. Under orders, select **Transfuse E-blood**



2. Clinician to acknowledge risk of haemolysis with uncrossmatched blood on EPIC during blood transfusion orders. During EPIC downtime, the physical form must be signed.

A screenshot of the EPIC form for acknowledging risk of haemolysis with uncrossmatched blood. The form has a blue header with 'Accept' and 'Cancel' buttons. The 'Transfuse:' section shows '1 UNIT' selected. Below this, there are two red warning icons. The first icon is followed by the text: 'Due to the critical condition of this patient, I request the immediate release of blood products for emergency transfusion ("Emergency Transfusion Supply").' The second icon is followed by the text: 'I understand the implications of and increased risks[i] incurred by the lack of complete compatibility testing.' Both of these text blocks are highlighted with a blue box. Below the second text block, there is a section for 'Comments' with a '+ Add Comments' button. The 'Process Instructions' section contains the following text: '(Routine testing of the Emergency Transfusion Supply will be completed as soon as possible and the attending physician will be notified of any abnormal results). 1 Risks of acute and delayed haemolytic transfusion reaction due to clinically significant red cell alloantibodies and/or autoantibodies. Frequency of red cell antibodies in the population is estimated to be 0.2% to 0.6%.' The form ends with 'Accept' and 'Cancel' buttons.

EPIC Downtime urgent transfusion acknowledgement form:

<https://nuh-intranet/Departments/Department-of-Laboratory-Medicine/Lab-eForms/Documents/BTS/G013%20Urgent%20Request%20for%20Blood%20and%20Blood%20Components.pdf>



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DEPARTMENT OF LABORATORY MEDICINE
BLOOD TRANSFUSION SERVICE
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**URGENT REQUEST FOR BLOOD AND BLOOD COMPONENTS
(Incomplete compatibility test)**

Patient's Name: _____ NRIC: _____

Date: _____ Time: _____ Location: _____

Sex: Male/Female Race: _____ Age: _____

Due to the critical condition of this patient, I request the immediate release of blood products for emergency transfusion ("**Emergency Transfusion Supply**").

I acknowledge that the Emergency Transfusion Supply has not undergone complete compatibility testing.

I understand the implications of and increased risks¹ incurred by the lack of complete compatibility testing. I acknowledge that the Blood Transfusion Service cannot be held liable for any complications or problems that may arise as a result of using the Emergency Transfusion Supply.

(Routine testing of the Emergency Transfusion Supply will be completed as soon as possible and the attending physician will be notified of any abnormal results).

Physician's Name: _____ Physician's Signature: _____

Physician's MCR No: _____

Tick the type of Emergency Transfusion Supply requested for:

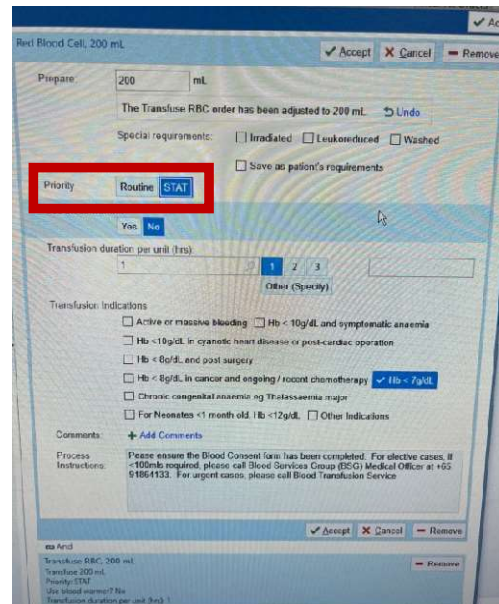
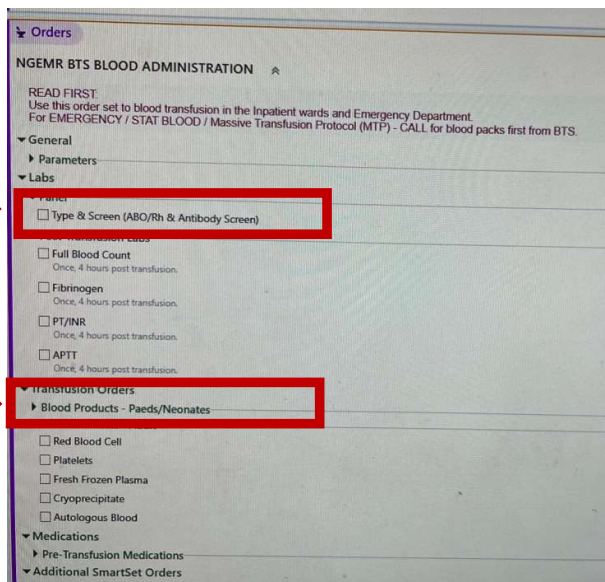
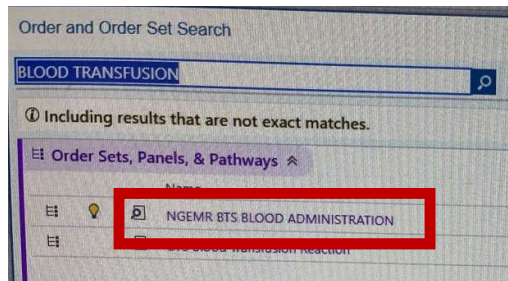
- ☐ Uncrossmatched Blood
- ☐ Rapid Crossmatch Blood (ABO Blood Group Compatible **ONLY**)
- ☐ Incompatible Crossmatched Blood due to red cell antibody (with consultation of Blood Bank Haematologist)

¹ Risks of acute and delayed haemolytic transfusion reaction due to clinically significant red cell alloantibodies and/or autoantibodies. Frequency of red cell antibodies in the population is estimated to be 0.2% to 0.6%.

Appendix II– How to order transfusion for rapid crossmatched/full crossmatched blood (from blood bank)

For rapid crossmatched:

1. Under orders, type **NGEMR BTS BLOOD ADMINISTRATION**
2. Click on Type & Screen
3. Click on Blood products – Paeds/Neonates
4. Fill in the details required. Click STAT for rapid crossmatched blood; routine for full crossmatched blood



Written by Dr Ivy Ang (Protocol w.e.f. 1 April 2026)

Drafted in collaboration with: Adult EMD Dr Kanwar SL, NC Siti Rabaah; Blood bank: Ms Karen Lim; Dr Lee Shir Ying; CE Dr Jasmine Ho, NM Thilagavathy Periyasamy. Reviewed by: Paediatric haematologist Dr Koh Pei Lin

Appendix III – Return of blood product form

[https://nuh-intranet/Departments/Department-of-Laboratory-Medicine/Lab-eForms/Documents/BTS/G026%20RETURN%20OF%20BLOOD%20PRODUCT%20FORM%20\(R5\).pdf](https://nuh-intranet/Departments/Department-of-Laboratory-Medicine/Lab-eForms/Documents/BTS/G026%20RETURN%20OF%20BLOOD%20PRODUCT%20FORM%20(R5).pdf)



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**DEPARTMENT OF LABORATORY MEDICINE
BLOOD TRANSFUSION SERVICE**

5 Lower Kent Ridge Road Singapore 119074 Tel: 6772 4344 Fax: 6772 4434

Return of Blood Product(s) Form

PART I: PATIENT PARTICULARS

Patient's Name: _____	Patient's Diagnosis: _____
NRIC/HRN: _____	Type of Operation (if any): _____
(Affix addressograph if any)	Ward/OT: _____

PART II: BLOOD PRODUCT RETURNED

Blood Product(s)	Unit Number(s)
_____	_____
_____	_____
_____	_____
_____	_____

PART III: REASONS FOR RETURNING PRODUCTS (Please specify in details)

- ☐ Not required: Please specify _____
- ☐ Transfusion delayed/ cancelled due to:
- ☐ Fever
 - ☐ Difficult cannulation
 - ☐ Others: _____
- ☐ Blood product integrity:
- ☐ Blood clot
 - ☐ Leakage (Location & cause of leakage: _____)
 - ☐ Others: _____
- ☐ Others: Please specify _____
- _____
- _____

PART IV: STORAGE CONDITIONS

Blood was kept in:

- ☐ cooler box with ice
- ☐ cooler box without ice
- ☐ OT/ EMD Refrigerator

Blood was returned in:

- ☐ cooler box with ice
- ☐ cooler box without ice**

****Note: Kindly return the blood unit ASAP**

Doctor's Name

Date/Time (hrs)

Consultant's Name

For Official Use Only:

Date/Time Returned: _____ Temp of Blood Product (°C): _____

Staff Remark(s): _____

DLM-FORM-GEN-026

R5-09-24

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Appendix IV – Blood bank number / location

Blood bank number – 22305

Blood bank – located on Main Building level 3, between Resident's lounge and ward 33.

